

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)
Vellore Institute of Technology, Chennai *Expected Graduation: 2026*

Experience

Research Intern May 2025 – July 2025
Centre for e-Automation Technologies (CeAT), VIT Chennai

- Developed a **CloudSim Plus** framework to evaluate the **Hippopotamus Optimization (HO)** algorithm for **VM placement** in cloud data centers.
- Enhanced simulation scalability by modularizing components, enabling reproducible testing across multiple datacenter configurations — GitHub: Repo

Full Stack Developer Intern Dec 2024 – Feb 2025
Daira Edtech Pvt Limited

- Contributed to the robust backend infrastructure by developing features and RESTful APIs for the Vidhira EdTech platform.
- Ensured data structure integrity and scalability leading to 30% improvement in data retrieval efficiency by designing effective data models to support new features on the Vidhira platform.

Web Development Intern Sept 2024 - Oct 2024
Indian Institute of Technology Bombay (IIT Bombay)

- Reduced data payload size by 90% for low-bandwidth environments by replacing text messages with numerical status codes in JSON responses.
- Enhanced backend robustness and API reliability by implementing Joi schema validation for the SaaS platform.

Projects

Water Brakes Aug 2024 - Nov 2024
Developed a Streamlit app that analyzes contour map data to recommend precise swale and trench locations for water management.
Enabled data-driven groundwater recharge decisions by processing UTM data to generate interactive Plotly visualizations.
Technologies Used: Streamlit, Python, Plotly, Matplotlib — Live: Demo

High-Performance Parallel AES Encryption Framework Jan 2025 - Apr 2025
Reduced encryption time by 51% on large payloads by engineering a parallelized AES system using Python’s ProcessPoolExecutor.
Increased throughput by 21.9% (to 362 KB/sec) compared to standard AES by designing a multi-core CTR-mode encryption pipeline.
Achieved a 49.95% avalanche effect, confirming cryptographic robustness and validating the dynamic S-box security approach.
Technologies Used: Python, Multiprocessing, SHA-512, Matplotlib, NumPy

Hybrid Ensemble Learning for Early Abiotic Stress Detection in Crops Aug 2025 – Nov 2025
Developed a hybrid ensemble pipeline combining an LSTM with additive attention and XGBoost for early-stage abiotic stress detection using 30-day physiological sensor time-series data.
Achieved **99.78% accuracy** and a **0.954 Macro F1 score** by learning robust temporal representations that capture dynamic plant stress patterns.
Improved performance on limited datasets using an **8-timestep sliding window**, generating **64-dimensional temporal embeddings** that enhanced model interpretability and early detection reliability.
Technologies: Python, LSTM, Attention Mechanisms, XGBoost, Scikit-learn, Pandas, Time-Series Analysis

Skills

- Programming Languages:** Python, C++, C, Java, HTML, CSS, SQL, JavaScript, Bash.
- Databases & Cloud:** MongoDB, PostgreSQL, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, Linux.
- Core Concepts:** Data Structures & Algorithms, DBMS, CN, OS, Cryptography (SHA-512/AES), Parallel Processing (Multiprocessing), System Design, CI/CD, System Optimization.

Positions of Responsibility

- Outreach Lead, DAO Community:**
 - Drove 50% community membership growth within 6 months by leading strategic outreach initiatives and enhancing engagement through numerous on-campus workshops and sessions facilitated via collaborations with leading Web3 companies (including Spheron, Polygon, blockOn, Huddle01, zkwallet).
 - Established the DAO as a key partner in the national/global Web3 ecosystem by securing alliances for 15+ nationwide hackathons and the TOKEN2049 Singapore conference, and playing a key organizing role in 5 major hackathons (250 participants each); notably led a 35-member team for Defy24’s record-breaking sponsorship and profit.
- Blockchain Lead, Hackclub:**
 - Expanded the blockchain department by 300% in 6 months by recruiting and actively mentoring over 25 junior students.
 - Developed members’ technical expertise by designing and delivering online blockchain workshops, creating learning roadmaps, and providing personalized skill development guidance.